

A Reproducible Tutorial on Synthetic-Data Scaling and Functional Collapse

Design, Scope, and Research Use of the Companion Notebook

Guillaume Harbonnier

Independent research note - MMALS / STRAT-Q programme

Version 1.0.2 - 16 June 2026

Abstract

This note documents the design, mathematical content, reproducibility boundaries, and intended research use of the notebook `synthetic_data_scaling_tutorial.ipynb`. The notebook accompanies the broader report *Synthetic Data in the Age of Scaling*, itself an independent analysis of Julia Kempe’s public lecture on synthetic data and model collapse. The notebook is deliberately lightweight: it reconstructs central mechanisms using transparent NumPy, pandas, and Matplotlib code rather than attempting to reproduce large language model fine-tuning, WikiText-103 experiments, or the full kernel-regression theory. Its modules illustrate Zipf-distributed long tails, Hutter’s infinite-memory missing mass, chopped-tail error floors, real-synthetic mixing, recursive pseudo-label drift, verifier enrichment, and a conceptual analogy with route collapse in MMALS. The principal contribution of the notebook is methodological rather than empirical: it separates executable demonstrations, formula-driven illustrations, conceptual hypotheses, and future validation requirements. This note makes those boundaries explicit and proposes the next steps needed to turn the tutorial into an evidence-producing MMALS experiment.

Status. This is a companion technical note for a pedagogical notebook. It is not a claim of reproducing the original large-scale experiments, not an empirical validation of MMALS, and not an official summary of Julia Kempe’s lecture.

Contents

1 Purpose and Position in the Repository	2
2 Notebook Architecture	2
3 Long-Tailed Data and Missing Mass	2
3.1 Zipf-like support	2
3.2 Scaling without abstraction	3
4 Synthetic Support Loss	3
4.1 Chopped-tail error floors	3
4.2 When synthetic data can help	4
5 Recursive Targets and Verification	4
5.1 Pseudo-label drift	4
5.2 Verifier enrichment	5
6 MMALS Research Bridge	5
7 Reproducibility and Auditability	6
8 From Tutorial to Evidence-Producing Experiment	7
9 Limitations	7
10 Conclusion	7

1 Purpose and Position in the Repository

The notebook serves as the executable bridge between three layers of the repository:

1. the lecture analysis and cited theory;
2. transparent mathematical illustrations that run on a laptop or Colab;
3. testable hypotheses for MMALS, STRAT-Q, and Test Data Governance.

Its primary design requirement is inspectability. Each mechanism is represented by a short code cell whose assumptions are visible. This choice trades empirical scale for conceptual clarity. The notebook therefore answers: *what should an engineer observe if the proposed mechanism is present?* It does not yet answer: *does a specific production model or MMALS release exhibit the mechanism?*

The source notebook is located at:

`notebooks/synthetic_data_scaling_tutorial.ipynb`

The broader lecture-analysis report cites the original video and the theoretical literature on scaling, long tails, model collapse, grokking, and verified synthetic data (Kempe, 2024; Hutter, 2021; Dohmatob et al., 2024a,b; Feng et al., 2025).

2 Notebook Architecture

The notebook contains seven executable modules followed by a research roadmap. Table 1 maps each module to its mathematical object and evidential status.

Table 1: Notebook modules and claim status.

Module	What is computed	Status
Zipf tail	Finite discrete distribution $p_i \propto i^{-\beta}$	Executable construction
Hutter missing mass	$E_T = \sum_i p_i (1 - p_i)^T$	Executable exact finite-support calculation
Chopped tail	$E(T, k) = T^{-c} + k^{-(\beta-1)}$	Formula-driven illustration
Real-synthetic mix	$(T_{\text{real}} + T_{\text{AI}})^{-c} + k^{-(\beta-1)}$	Formula-driven illustration
Recursive relabeling	Hand-specified stochastic drift of slope and intercept	Conceptual simulation
Verifier enrichment	Bayes precision among accepted candidates	Executable analytical calculation
MMALS host ecology	Scripted host route shares under recursive and anchored regimes	Conceptual hypothesis only

This classification is important. A plotted curve can be reproducible without being empirical evidence for the phenomenon in a neural network. Reproducibility means that the same assumptions produce the same result; validation means that observed data support those assumptions in the target system.

3 Long-Tailed Data and Missing Mass

3.1 Zipf-like support

The notebook constructs 5,000 ranked events with

$$p_i = \frac{i^{-\beta}}{\sum_{j=1}^{5000} j^{-\beta}}, \quad \beta = 1.5. \quad (1)$$

For this finite distribution, approximately 93.37% of the mass lies in the first 100 ranks, while approximately 1.35% remains after rank 1,000. The latter mass is small in aggregate but may represent thousands of distinct rare words, incidents, skills, or routes.

3.2 Scaling without abstraction

The Hutter-style learner memorizes every observed event and fails only when the test event has not appeared in the training sample. Its expected error is

$$E_T = \sum_i p_i (1 - p_i)^T. \quad (2)$$

The notebook evaluates this expression directly for a range of T . Under a power-law distribution, its asymptotic behavior is

$$E_T \propto T^{-(\beta-1)/\beta}. \quad (3)$$

For $\beta = 1.5$, the reference exponent is $1/3$. This is the notebook's first methodological warning: a clean power-law learning curve can arise from coverage by exact memory, without abstraction or compositional generalization (Hutter, 2021).

Non-claim. The notebook does not infer a scaling exponent from an LLM experiment. It evaluates a finite probabilistic model and overlays the corresponding theoretical reference slope.

4 Synthetic Support Loss

4.1 Chopped-tail error floors

If synthetic generation retains only ranks up to k , the illustrative law is

$$E(T, k) \asymp T^{-c} + k^{-(\beta-1)}, \quad c = \frac{\beta-1}{\beta}. \quad (4)$$

The first term decreases with sample count. The second term is independent of sample count and represents mass missing from the generated support. Figure 1 shows the resulting plateaus.

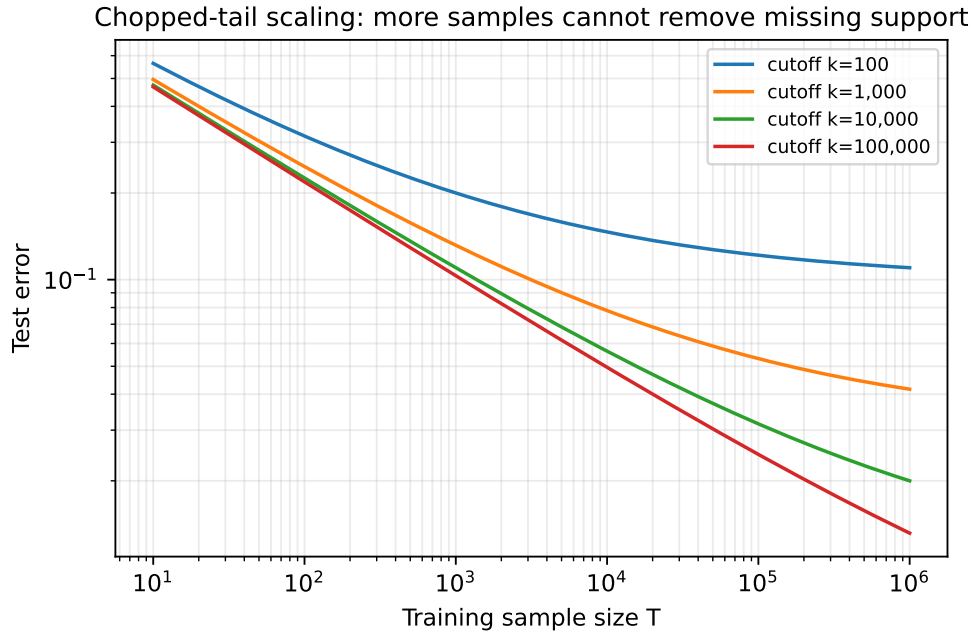


Figure 1: Formula-driven reconstruction of chopped-tail scaling. Increasing T cannot remove a support-dependent floor.

This cell is intentionally simple: it plots the asymptotic expression rather than training a generator. It is useful for reasoning about failure modes, but it does not estimate k for a real model.

4.2 When synthetic data can help

The real-synthetic mixing cell uses

$$E(T_{\text{real}}, T_{\text{AI}}, k) = (T_{\text{real}} + T_{\text{AI}})^{-c} + k^{-(\beta-1)}. \quad (5)$$

The figure illustrates that synthetic samples can reduce finite-sample error when real data are scarce, while the support floor remains unchanged.

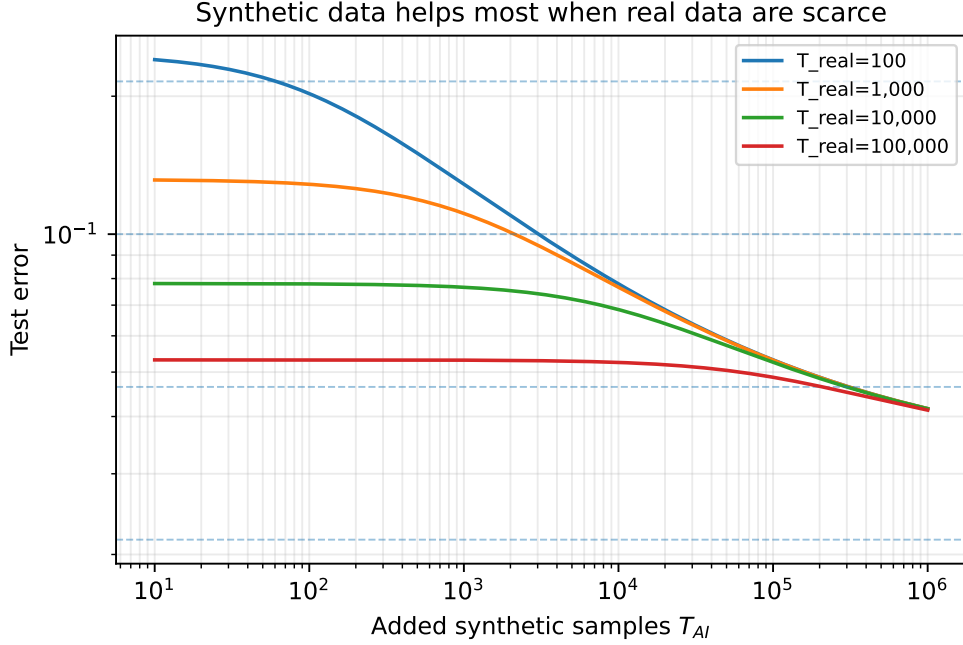


Figure 2: Illustrative benefit of synthetic data under real-data scarcity, with an unchanged missing-tail floor.

The operational interpretation is precise: synthetic data can densify a known region of the problem but does not automatically enlarge the set of represented regimes.

5 Recursive Targets and Verification

5.1 Pseudo-label drift

The notebook represents a true linear relation by (w_0, b_0) and recursively perturbs the previous generation's slope and intercept. This is not a fitted ridge-regression experiment. It is a visual surrogate for the principle

$$w_0 \rightarrow \hat{w}_1 \rightarrow \hat{w}_2 \rightarrow \dots, \quad (6)$$

where each model becomes the next model's labeling source. The full theoretical treatment of recursive regression separates bias, variance, noise, regularization, and generational depth (Dohmatob et al., 2024b). The notebook merely makes the direction of drift visible.

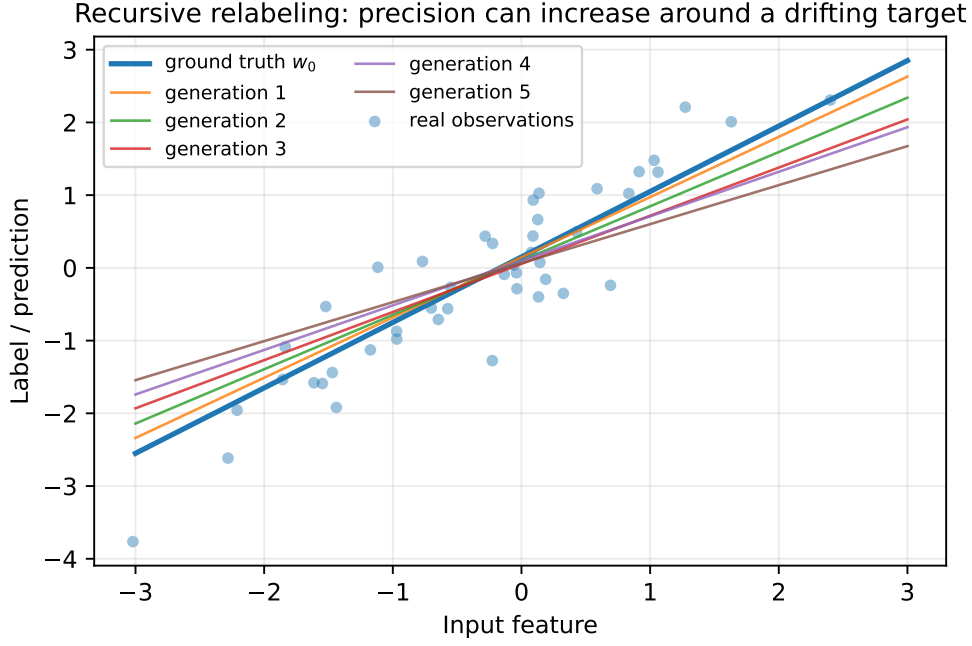


Figure 3: Conceptual recursive pseudo-label drift. The plotted coefficients are illustrative rather than estimated from data.

5.2 Verifier enrichment

Let a generator be correct with probability q . Let a verifier accept a correct candidate with probability ϕ and accept an incorrect candidate with probability ψ . Bayes' rule gives

$$\mathbb{P}(\text{correct} \mid \text{accepted}) = \frac{\phi q}{\phi q + \psi(1 - q)}. \quad (7)$$

For $q = 0.20$, the notebook reports the accepted-set precisions in [Table 2](#).

Table 2: Verifier enrichment calculated by the notebook.

ϕ	ψ	Accepted-set correctness
0.95	0.05	82.61%
0.80	0.10	66.67%
0.60	0.20	42.86%
0.90	0.30	42.86%

The final two rows show why acceptance rate alone is insufficient. A verifier that accepts many correct candidates can still be weak if it also accepts too many incorrect ones. This module operationalizes the generate-verify-select-retrain principle studied in verified synthetic-data bootstrapping ([Feng et al., 2025](#)).

6 MMALS Research Bridge

The final figure is a deliberately scripted analogy. In the recursive-only scenario, host H_1 progressively dominates while H_2 and H_3 vanish. In the anchored scenario, independent evidence preserves a broader ecology.

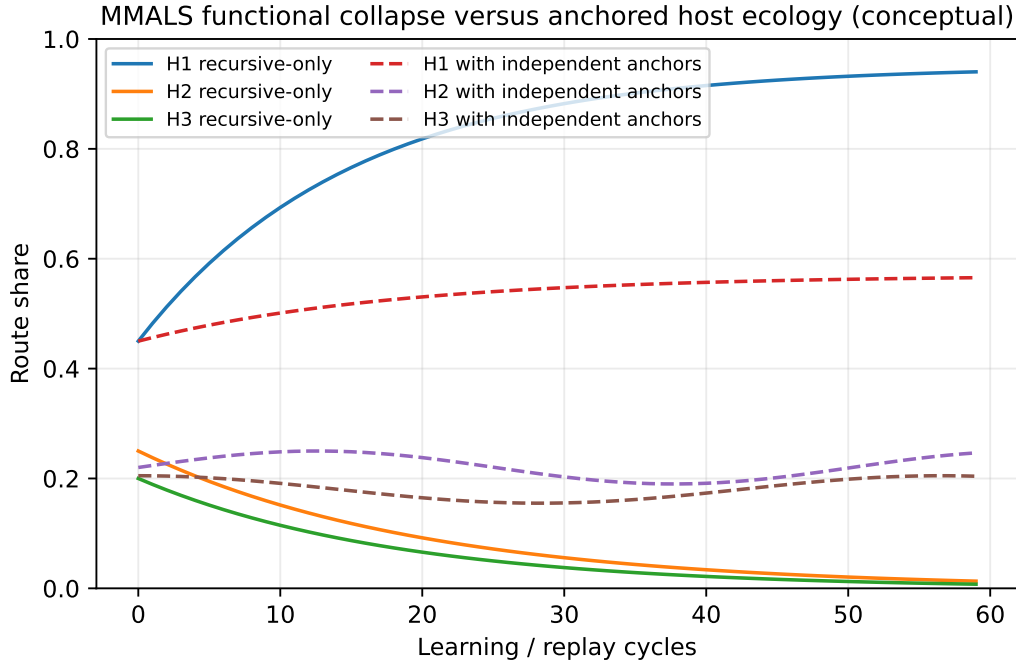


Figure 4: Conceptual MMALS route-collapse hypothesis. This figure is not produced from MMALS traces.

The analogy maps synthetic-data collapse to functional collapse:

$$\text{frequent modes} \leftrightarrow \text{dominant host or route}, \quad (8)$$

$$\text{lost tail} \leftrightarrow \text{lost specialized functions}, \quad (9)$$

$$\text{recursive pseudo-labels} \leftrightarrow \text{self-reinforcing replay traces}. \quad (10)$$

The notebook therefore formulates, but does not validate, the following hypothesis:

A routing ecosystem may become locally stable while globally losing accessible, causally useful functions if its memory and replay are dominated by its own previously selected routes.

A genuine MMALS experiment must replace scripted shares with observed route traces and must distinguish harmless concentration on simple cases from irreversible loss of functional support.

7 Reproducibility and Auditability

The notebook uses only NumPy, pandas, and Matplotlib. A fixed random generator seed, 42, makes the stochastic relabeling illustration reproducible. The deterministic modules reproduce exactly, modulo library and rendering differences.

A reproducibility claim for this notebook therefore means:

- formulas, parameters, and random seeds are visible;
- each figure can be regenerated locally;
- conceptual figures are labelled as conceptual;
- no downloaded model weights or hidden API calls are required;
- assumptions can be modified cell by cell.

It does not mean:

- exact replication of WikiText-103 fine-tuning;
- replication of published Llama experiments;
- derivation of the full kernel-ridge theorems;
- evidence that MMALS currently collapses or resists collapse.

8 From Tutorial to Evidence-Producing Experiment

The next implementation stage should create a trace-driven notebook with four controlled protocols:

1. **Independent-only:** training and evaluation from externally validated traces;
2. **Recursive-only:** new route labels produced by the previous MMALS generation;
3. **Fixed anchor:** recursive traces mixed with a fixed real or independently verified anchor;
4. **Fresh anchor and verified synthetic:** new independent traces plus synthetic candidates accepted by an external oracle.

The minimum outputs should include:

- route distribution and route entropy by regime;
- functional tail coverage;
- host contribution gain and ablation impact;
- route-swap and candidate-permutation sensitivity;
- performance on exact repeats, nearby variants, and new compositions;
- generation depth, source lineage, freshness, and verifier identity;
- compute cost and retained-memory cost;
- multi-seed uncertainty.

A route should not be preserved merely because it is rare. It should be preserved when it is rare, causally useful, and relevant to a real or stress-tested regime.

9 Limitations

The notebook intentionally omits several complexities:

1. finite-support normalization slightly modifies ideal asymptotics;
2. the chopped-tail cells omit unknown multiplicative constants;
3. pseudo-label drift is scripted rather than estimated;
4. the verifier module assumes known q , ϕ , and ψ ;
5. verifier-induced selection bias is not simulated;
6. no autoregressive sequence model is trained;
7. no grokking or late transition is measured;
8. no real MMALS host, router, memory, or objective is executed.

These omissions are not defects relative to the notebook's purpose. They define the boundary between an explanatory tutorial and a research experiment.

10 Conclusion

The companion notebook is best understood as a compact executable argument. It shows that long-tailed data can generate scaling without abstraction, that missing support creates error floors, that synthetic data can help when real data are scarce, that recursive labels can drift, and that a discriminating verifier can enrich a weak generator's outputs. It then translates these mechanisms into a falsifiable MMALS hypothesis about host and route collapse.

Its scientific value lies in disciplined scope. The notebook does not claim that a toy curve is evidence about a production LLM or MMALS release. Instead, it provides a common language, a set of expected signatures, and a precise plan for replacing illustrative curves with real traces, independent anchors, causal interventions, and auditable evidence.

References

- Julia Kempe. *Synthetic Data – Friend or Foe in the Age of Scaling?* Public lecture in the IHES Mathematics for and by Large Language Models series, 2024. https://www.youtube.com/watch?v=D9x3DT16mGg&list=PLx5f8IelFRgHrJ9W6_fbf03ahDrXMEIWn.
- Marcus Hutter. Learning curve theory. *arXiv preprint arXiv:2102.04074*, 2021.
- Elvis Dohmatob, Yunzhen Feng, Pu Yang, Francois Charton, and Julia Kempe. A tale of tails: Model collapse as a change of scaling laws. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Elvis Dohmatob, Yunzhen Feng, and Julia Kempe. Model collapse demystified: The case of regression. *arXiv preprint arXiv:2402.07712*, 2024.
- Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. Beyond model collapse: Scaling up with synthesized data requires verification. In *International Conference on Learning Representations*, 2025.